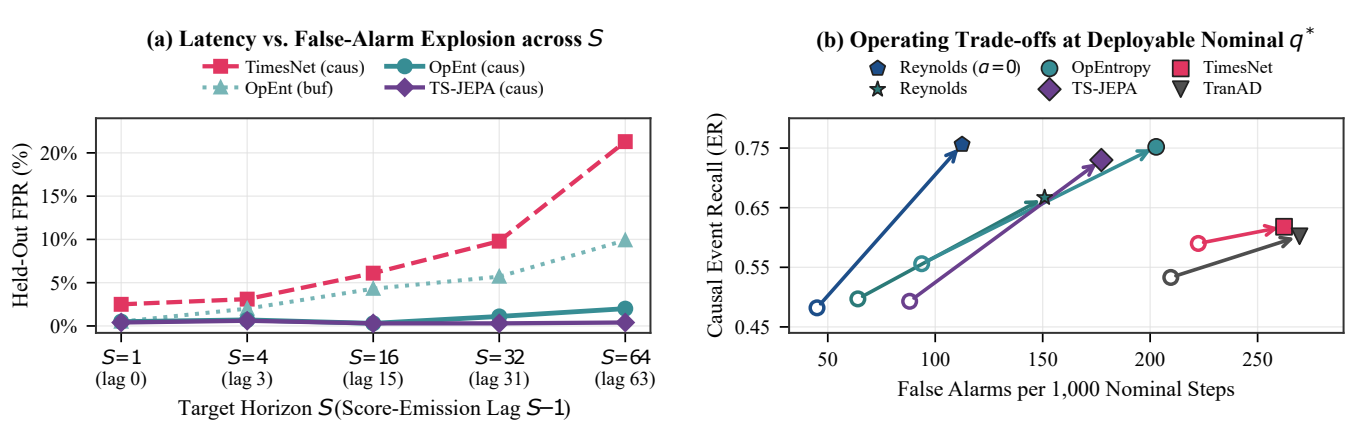

(a) Latency vs. False-Alarm Explosion across S
TimesNet (caus)
OpEnt (caus)
OpEnt (buf)
TS-JEPA (caus)
Held-Out FPR (%)
20%
15%
10%
5%
0%
$S=1$ (lag 0)
$S=4$ (lag 3)
$S=16$ (lag 15)
$S=32$ (lag 31)
$S=64$ (lag 63)
Target Horizon S (Score-Emission Lag $S-1$)
(b) Operating Trade-offs at Deployable Nominal q^*
Reynolds ($\alpha=0$)
Reynolds
OpEntropy
TS-JEPA
TimesNet
TranAD
Causal Event Recall (ER)
0.75
0.65
0.55
0.45
50
100
150
200
250
False Alarms per 1,000 Nominal Steps